Current Sustainable/Renewable Energy Reports (2023) 10:206–217
https://doi.org/10.1007/s40518-023-00224-3

Improving the Representation of Climate Risks in Long-Term Electricity Systems Planning: a Critical Review

James Doss-Gollin[1] · Yash Amonkar[2] · Katlyn Schmeltzer[3] · Daniel Cohan[1]

Accepted: 31 July 2023 / Published online: 16 August 2023
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2023

Abstract

Purpose of Review Electricity systems face substantial climate risks which are escalating due to electrification, renewable energy intermittency, population changes, and the intensifying impacts of climate change such as extreme temperatures and weather-induced infrastructure damage. This critical review investigates climate risks to the electricity sector and scrutinizes the methodologies used to represent climate risk in long-term electricity system planning studies.

Recent Findings Climate risks to electricity systems are driven by extreme weather, average weather, technology, and other social and technological factors. All are expected to evolve in the future. Future climate risks to electricity systems depend on interactions between each, and thus assessing future climate risks to electricity systems requires exploring a wide range of possible futures.

Summary Many studies rely on weather data and socio-economic scenarios that are inadequate to fully characterize climate risks to present and future electricity systems. We advocate for more holistic assessments that incorporate comprehensive weather data, acknowledge dynamic multi-sector interactions, and employ adaptive and robust methodologies.

Keywords Climate risk · Electricity systems · Climate change

Introduction

The impacts of recent weather extremes worldwide highlight the vulnerability of electricity systems to severe weather. This is particularly true when extreme temperatures cause spikes in heating or cooling demand while disrupting power supply, resulting in challenges to meet peak loads on electricity systems. In 2022, for example, heatwaves in India and Pakistan caused unprecedented spikes in electricity demand that exceeded supply [1]. Similarly, during a severe freeze in Texas in February 2021, record-breaking demand for heating coincided with supply failures, leaving millions without power [2]. Severe weather also affects energy supply by limiting thermal and renewable generation and damaging transmission and distribution infrastructure. For example, power shortfalls that exacerbated the impacts of the 2021 Texas freeze were caused by failures throughout the electricity supply chain, especially in natural gas fuel supply and power plant operations [3]. From the perspective of human health, the vulnerability of electricity systems to tropical cyclones (TCs) [4, 5] and severe weather [6] is a particularly critical challenge because electricity is often most needed during severe weather. For example, it is estimated that a concurrent blackout and heatwave event could kill 1% of the population of Phoenix, AZ, USA [7].

Factors beyond extreme weather also contribute to growing climate risks for electricity systems. For example, population growth, electrification, and economic development are driving increases in electricity demand [8]. At the same time, the increasing penetration of intermittent renewable genera-

✉ James Doss-Gollin
jdossgollin@rice.edu

Yash Amonkar
amonkar@unc.edu

Katlyn Schmeltzer
kes20@rice.edu

Daniel Cohan
cohan@rice.edu

[1] Department of Civil and Environmental Engineering, Rice University, 6100 Main Street, Houston, TX 77005, USA

[2] Center on Financial Risk in Environmental Systems, UNC Institute for the Environment, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[3] Department of Chemistry, Rice University, 6100 Main Street, Houston, TX 77005, USA

tion, motivated in part by commitment to mitigating global climate change [9], is changing the electricity system's vulnerability to weather risks. Studies that examine the influence of climate change on existing infrastructure find that while overall changes in demand for electricity may be modest, warmer temperatures will drive large increases in summertime peak demands [10–12].

In light of these evolving risks, the primary objective of this paper is to critically review the methods used to incorporate climate risk into long-term energy systems planning. In this paper, we focus on a key question: are these methods sufficient to accurately assess and robustly manage climate risks? Previous reviews have addressed parts of this literature, including methods for data compression and slicing [13], wind resources in a changing climate [14], community resilience [15], and climate change impacts on energy demand and generation potential [16–18]. Our contribution is to apply a multisector, climate risk perspective with a particular emphasis on comparing best practices identified in the field of climate risk management to identify research gaps and opportunities for the electricity sector. We structure the paper as follows: In the "Vulnerabilities of electricity systems to extreme weather" section, we review vulnerabilities of electricity systems to extreme weather, considering impacts on demand, supply, and transmission and distribution. In the "Evolving climate risks to electricity systems" section, we consider how these vulnerabilities may evolve, summarizing the expected effects of climate change on extreme weather, how these effects have been projected to affect electricity systems, and how technological and social change will affect overall climate risk. In the "Representation of climate risks in long-term planning studies" section, we summarize how climate risks are represented in long-term electricity planning models and analyses, which we contrast in the "Research gaps and opportunities" section with the state of the art to identify three focus areas for improvement. We conclude in the "Summary" section with a brief summary.

Vulnerabilities of Electricity Systems to Extreme Weather

We begin by identifying vulnerabilities of electricity systems to extreme weather. In the context of energy systems, vulnerability refers to the susceptibility of the system to suffer damage or disruption due to external factors, such as weather events. Vulnerability is related to reliability, resilience, and robustness. Reliability, as defined in power systems, is the probability of a normal operation of the electrical grid at a given time, ensuring a continuous supply of energy by measuring the frequency, duration, and scale of supply interruptions. Resilience is the ability to limit the extent, severity, and duration of system degradation following an extreme event. Robustness, on the other hand, is the inherent ability of the electricity system to endure a set of disturbances and maintain functionality, performing well over a wide range of plausible futures. Understanding how weather affects these aspects is critical to safeguarding electricity systems.

Weather Impacts on Electricity Demand

Temperature is the primary weather variable that influences energy demand, although other variables like humidity also matter [19]. A deep literature characterizes the temperature-electricity demand relationship using a variety of statistical methods [20, 21]. For example, analysis of different studies modeling electricity demand in Texas using a polynomial fit [22], a binning method [23], and a tree-based machine learning approach [24] finds that these approaches yield similar estimates under normal operating conditions, yet diverge substantially under extreme conditions for which impacts are especially great.

Weather Impacts on Electricity Generation

Extreme weather and normal weather fluctuations affect all forms of electricity generation.

For example, the output of solar photovoltaic panels is primarily dependent on the amount of sunlight they receive and the ambient air temperature [25–27]. Wind power is highly sensitive to variations in wind speed: wind power scales cubically with wind speed, but turbines also have a minimum wind speed required to start generating power ("cut-in"), a speed at which peak mechanical output is reached, and a speed at which turbines must be shut down for safety ("cut-out") [25]. Furthermore, local topography, turbine design, and wake effects can significantly impact wind generation [28].

Hydropower generation is dependent on water availability, and drought conditions can limit power production by reducing water levels and increasing evaporation rates. Additionally, during hot or dry conditions, water availability may be further limited due to increased demand from urban and agricultural sectors [29–32].

Thermal power plants, including coal, natural gas, and nuclear plants, rely on air and water for cooling. The efficiency of these plants decreases with rising temperatures [8, 33]. Hot weather can lead to reductions in thermal power generation, and this issue is compounded by droughts which limit the availability of cooling water [30, 34, 35]. Cold temperatures can also affect production, transport, and generation of thermal power electricity when facilities are not fully weatherized [3].

Climate variability at multiple spatial and temporal scales can also stress electricity systems, even if the conditions are not extreme at a particular moment. For example, in

northern Europe, periods of low wind and sunlight, termed "dunkelflaute," can persist for weeks and are linked to persistent or quasi-stationary weather patterns called blocks [36, 37]. Similar patterns have been observed in other regions including Western North America, Australia, and Japan [38–40]. A range of other climate oscillations that drive weather risks at subseasonal to multidecadal time scales are documented in the climate literature [41, 42]. For example, the North Atlantic Oscillation is an important driver of supply and demand fluctuations in Europe, especially from late winter to early spring [43–45].

Weather Impacts on Electricity Transmission and Distribution

Extreme weather poses significant challenges to the transmission and distribution of electricity, making it a critical factor in climate risk assessment. Flooding can threaten electrical systems by inundating substations and underground cables, causing temporary outages or long-term damage [18]. High temperatures can reduce electric transmission capacity by affecting the thermal properties of transmission lines [46]. Snow and ice can accumulate on overhead power lines, straining them beyond their mechanical limits and leading to collapses and cascading outages [18, 47–49]. Wildfires are particularly hazardous to electrical systems in arid regions, as they can damage power lines and other infrastructure [18, 50]. Finally, severe storms, such as TCs, can inflict extensive damage on transmission and distribution infrastructure, primarily through fallen trees and broken transmission lines. These impacts can be devastating, leading to prolonged outages and requiring substantial resources for restoration [51, 52].

Evolving Climate Risks to Electricity Systems

Evolving climate risks to electricity systems stem from the interplay of climate changes, technological evolutions, and societal shifts. Notably, while climate change directly poses hazards to electricity supply and demand, technological and social alterations can modify how different weather conditions impact these systems.

Climate Change Impacts on Extreme Weather

Before examining evolving climate risks to electricity systems, we summarize consensus science on anticipated changes in weather patterns that are directly relevant to electricity systems.

Many forms of hazardous weather are expected to increase in frequency and intensity as the climate warms. For example, there is an expectation for an increase in frequency, intensity, and amount of heavy precipitation events as well as agricultural and ecological droughts in many regions [53]. While the sensitivity of TC frequency to warming remains uncertain [54, 55], increases in global average TC rain rates of about 12% for a 2° C global warming (relative to pre-industrial) are projected, consistent with the Clausius-Clapeyron scaling of saturation-specific humidity [54], and peak rainfall rates are likely to increase faster than this for some regions [53]. Moreover, global maximum TC surface wind speeds are projected to increase by about 5% at 2°C [54]. Considering projected sea level rise [56] and TC dynamics, risk from coastal storm surge is projected to increase substantially [57]. Similarly, Convective Available Potential Energy, which describes the potential of the atmosphere to generate heavy rainfall and thunderstorms, is projected to increase in the tropics and subtropics, as is precipitation from convective storms [53]. Climate change is also likely to make wildfires more frequent and severe worldwide, with particular increases at high latitudes and in some mountainous regions [18, 58].

Changes in average and extreme temperatures are also expected. One important point is that warming over land will exceed global average increases in temperature [59]. Climate change also affects the diurnal and seasonal cycles: warmer and more frequent hot days and nights over most land areas are expected, with mid-latitude, semi-arid regions and the South American Monsoon region experiencing the most significant increases [53]. Additionally, in most subtropical and mid-latitude land regions, except some parts of Asia, models predict more pronounced warming during summer than winter, resulting in an amplified seasonal cycle [60, 61].

The effect of climate change on persistent weather regimes remains an area of active research. In the mid-latitudes, weather regimes are influenced by storm tracks, which are in turn affected by competing processes with uncertain net results [62]. There is a projection for an increase in the size of atmospheric blocks, particularly during summers in the Northern Hemisphere [63], but robust trends have only been identified in observations for some regions and seasons [64]. Physical mechanisms such as quasi-resonant amplification have been linked with climate extremes [65–67], bolstering evidence that global warming increases the persistence of summer weather [68], but the magnitude and extent of these changes remain uncertain.

Climate Change Impacts on Electricity Systems

Electricity Demand

A critical driver of changes in electricity consumption patterns is increased temperature, particularly due to the use of electrical heating and cooling systems. For example, researchers combined an empirical relationship between high temperatures and electricity demand for California with

projections of future temperature, holding technology and population constant, and found that electricity demand and deficits could surge [69]. Similarly, a comparison of 1996–2005 weather data in Texas with downscaled projections under RCP 8.5 for 2041–2050 revealed that under this high-emissions scenario, net demand could increase by up to 6% and thermal deratings during peak hours could increase by 40% [26]. A retrospective and temperature-based analysis observed consistent increases in summer average and peak cooling demand across the continental United States [12]. While warmer winters reduce total electricity demand during winters, researchers found that the likelihood and severity of extreme cold temperatures in the mid-latitudes, which are particularly consequential for electricity systems, are unlikely to decrease for several decades [2, 70].

Electricity Supply

Climate change affects electricity generation in various ways, specific to particular regions and technologies.

Climate change is not expected to substantially impact global solar insolation [18]. While regional variations in dimming and brightening driven by changes in cloud cover, aerosols, and water vapor are possible, these are not likely to significantly lower the global potential of solar energy. Wind power is more susceptible to the influences of climate change, and numerous studies have modeled wind production for different regions and climate scenarios. Most studies find overall reductions to be modest; for example, a regional climate model assessment of wind resources in Europe finds reductions of $< 5\%$ at 2°C of global warming for all countries considered [71], in line with other studies [72, 73], though in specific regions some studies find larger impacts plausible [74, 75].

Hydropower presents unique challenges and considerations. Generally, increased precipitation is expected to increase water availability and, consequently, hydropower production. However, increased precipitation intensity can affect dam structures and power production by increasing debris accumulation and vegetation growth [18]. Additionally, models predict that many snow-fed systems will experience a shift from snow to rain in winter and earlier snowmelt, which can shift the seasonality of hydropower production [76]. Finally, changes in the risk of interannual droughts are uncertain but could pose significant risks to hydropower systems in some regions [18]

Thermal power generation is also affected by climate change, particularly due to its dependence on water resources for cooling. In general, significant decreases in available thermal power plant capacity due to climate change are projected [16, 29]. Several options to mitigate the impacts of reduced cooling water availability, increased water temperature, and lower flows have been suggested, including switching from freshwater to seawater or air cooling, replacing once-through cooling systems with recirculation systems, and increasing the efficiency of power plants [77].

Finally, it is critical to consider the uncertainties and complexities associated with these projections. For instance, when examining least-cost options for the hydropower-dominated Ecuadorian power system up to 2050, the uncertainty between different Earth system models (ESMs) ("model uncertainty") was found to surpass the uncertainty between different climate scenarios ("scenario uncertainty") [78], consistent with other studies [e.g.,79, 80]. Uncertainties in renewable energy potential and demand could trigger a significant performance gap and drop in power supply reliability due to future climate variations and extreme weather events [81]. This emphasizes the need to adopt bottom-up risk assessment and management approaches to more effectively address the complexities and uncertainties in the energy sector.

Technological and Social Factors Contributing to Climate Risks for Electricity Systems

Energy system transformation is essential for climate change mitigation, but investments in the energy and electricity sectors respond to a wide range of concerns including economic development, energy access, energy justice, energy security, air pollution, land use, and energy cost [8, 18]. To understand climate risks to future electricity systems, it is critical to assess interactions between different social and technological pathways and climate change.

Despite efficiency gains, electricity consumption is projected to increase in most regions due to population growth and economic development [82]. In places ranging from Texas to West and Central Africa, this can increase the sensitivity of electricity demand to temperature [23, 83].

The integration of electricity into new sectors, including road transport, heating, industrial processes, and electrolytic hydrogen production, is anticipated to alter load curves and has the potential to make electricity demand more variable [8]. The electrification of heating is particularly significant. Currently, a large proportion of energy used for home heating is sourced from natural gas or oil, but a shift to electric heating is considered critical for the energy transition. Electric heat pumps are far more efficient than traditional electric furnaces overall, but their energy use increases more sharply as temperatures decline. One study accounting for temperature-dependent energy demand, local building stocks and heating technologies, population, and other socio-economic factors to calculate electricity demand in the USA finds that even with the most energy-efficient heat pumps, electrification of heating would cause wintertime peaks to be 70% higher than current summertime peaks overall and more than twice as high in 23 states [84].

A particular challenge is that in many regions, the inter-annual variability of peak heating demand (i.e., winter temperature extremes) is substantially larger than peak cooling demands (i.e., summer temperature extremes). As electrification of building heating continues, grids must prepare for rare and variable winter peaks rather than for frequent and more predictable summer peaks [2, 12].

Over the past decade, technological advances such as the falling costs of solar panels and lithium-ion batteries have transformed the energy sector [85]. While future technological advances are deeply uncertain [86, 87], understanding the impact of plausible technological pathways on climate risks to electricity systems is critical. For example, demand-side response schemes, including smart appliances and electric vehicles, can provide flexibility, while energy storage technologies can provide a variety of ancillary grid services [88].

Representation of Climate Risks in Long-Term Planning Studies

Given the significance of evolving climate risk to electricity systems, it is crucial to scrutinize how these risks are incorporated into the long-term electricity systems planning literature. In this section, we review several influential studies that focus on long-term energy planning with the objective of achieving a net-zero or clean electricity future. While this review is not exhaustive, these studies are broadly representative of the field.

The Princeton "Net-Zero America" study [89] investigates the least-cost energy mix necessary for reaching net-zero emissions by utilizing the EnergyPATHWAYS and RIO models. The National Renewable Electricity Laboratory (NREL) "Examining Supply-Side Options to Achieve 100% Clean Electricity by 2035" [90] focuses on identifying the energy mix and requisite grid modifications to realize 100% clean electricity by 2035, employing the ReEDS model. The International Energy Agency's "World Energy Outlook 2022" [8] offers a critical analysis of trends in energy demand and supply and the associated implications for energy security, environmental protection, and economic development. The Decarb America "Pathways to Net-Zero Emissions" study [91] concentrates on evaluating the trade-offs between different strategies aimed at net-zero emissions and gauging the economic opportunities that emerge from a clean energy economy. Finally, we consider two academic papers [92, 93] that employ a more simplified model to study the feasibility of achieving high renewable energy penetration and analyze the implications for storage and overbuilding.

Many studies use only a few years of weather data to assess climate risks, and only some impacts of weather on electricity systems are typically represented. For example, [89] samples electricity operations using 41 days from the 2011 weather year. The Pathways to Net-Zero Emissions study [91] uses heating and cooling degree days from 2000 and 2017 to model electricity demand. The NREL study [90] incorporates a more extensive dataset, with timeslice profiles based on the weather year of 2012 and hourly profiles from 7 years (2007–2013). The two academic studies considered [92, 93] utilize a broader dataset, consisting of 36 and 39 years, respectively, of hourly reanalysis data. Finally, the World Energy Outlook 2022 [8] models electricity demand and supply without relying explicitly on weather. None of these data sources considers climate change impacts, and the short data lengths used may not be sufficient for capturing the full range of weather variability [42].

With respect to non-weather variables, a majority of the studies explore only a handful of technological, social, and policy futures [8, 90, 91, 94]. The Priceton study [89] explores 55 scenarios representing deviations from the baseline scenario, providing some characterization of uncertainties, though this "one at a time" sensitivity analysis limits the conclusions that can robustly be drawn [95–97]. Shaner et al. [92] and [93] analyze a broad range of values for the carbon intensity of electricity. Generally, there is a need for more comprehensive scenario analyses to address the intricacies and uncertainties inherent in long-term energy planning.

Research Gaps and Opportunities

Accurately characterizing and quantifying climate risks to electricity systems requires not only drawing many samples from the distribution of possible weather but also considering the interaction with other pertinent factors that drive demand and vulnerability. In this section, we draw from literatures on climate risk management, including but not limited to electricity systems, to identify areas to strengthen the assessment of climate risks to electricity systems.

Representing Weather Extremes and Variability

Forecasting energy demand and generation in electricity systems necessitates high-resolution weather data, such as temperature, insolation, wind speed, and streamflow. However, acquiring such data is challenging, and while ESMs offer valuable insights, they face fundamental limitations in (i) simulating small-scale hazards [98, 99], (ii) representing some dynamical processes [100–104], and (iii) exploring the full range of climate emissions forcings [105]. At the same time, relying solely on observational data for risk assessment might not be indicative of future scenarios given the non-stationarity induced by climate change [42, 106] and sampling bias from using short observational records [107–110].

Downscaling methods aim to tackle the scale mismatch between coarse climate data and the site-specific information needed for energy planning. A challenge is that the statistical relationship between inputs (e.g., reanalysis fields) and outputs (e.g., site-scale weather variables) may not hold, either because the future inputs are different (e.g., if climate model outputs are used as input) or because climate change can alter the statistical relationship [111, 112]. Advanced techniques like generative computer vision show promise, but they require extensive training data and share the fundamental assumptions of classical downscaling methods [113, 114].

Bootstrap methods can be used to extend historical data to create scenarios conducive to risk management [115–117]. These methods have the advantage of reduced dependence on parametric statistical assumptions but are still constrained by the extent of historical data and the challenge of accounting for climate change-induced nonstationarity. Stochastic weather generators that combine bootstrap-like methods, physical models, nonparametric statistics, and parametric statistics have been used to generate synthetic meteorological sequences that capture the properties of the observed variables [118–121]. These models often outperform model chains that rely on downscaled and bias-corrected climate variables [122], but these approaches require significant customization for each location and application.

Managing the complexity and computational demands of high-resolution datasets and simulations in energy system optimization is another challenge. Data compression techniques like time-slice compression address this by strategically selecting and simulating representative time intervals from the dataset instead of the entire timeline [13]. For instance [89], selects 41 days per year, equivalent to 984 h, to sample electricity operations, significantly reducing computational costs. However, this approach has limitations, especially when dealing with stocks, like batteries and hydropower, where decisions made at one point can have nonlinear effects on future options.

Representing Multi-Sector Dynamics

Assessing climate risk in multisector systems necessitates a comprehensive approach that represents the complex interplay between climate and socioeconomic variables. Such assessments need to dynamically and endogenously account for interactions across sectors, rather than treating them as isolated boundary conditions or external factors [96]. Understanding the interconnected nature of these interactions and transparently communicating modeling choices are crucial for accurately representing societal systems on varying spatio-temporal scales [96]. For example, a study expanded the five Shared Socioeconomic Pathways by analyzing over 30,000 scenarios from an integrated assessment model and found that many of the most significant scenarios arose from interactions between different parameters [123]. Another study coupled an Earth system model, a global hydrologic model, and an economic surplus loss metric and concluded that impactful scenarios often arise from combinations of standard scenarios [124]. Furthermore, a study combined a damage model, flood frequency model, and economic model to evaluate the optimal elevation of the house under uncertainty and found that the flood probability distribution and the discount rate drive the outcomes [125]. These studies collectively underscore the imperative for integrated modeling and comprehensive scenario analysis in assessing climate risk in multi-sector systems.

In the energy sector, certain models have made strides in capturing these risks. For example, the impact of climate risk on human health through power plant emissions was examined using a 500-year synthetic weather ensemble, highlighting the highest damages during hot, dry years when hydropower availability dwindled and electricity demand surged, and implementing a tax on power plant operations based on their contribution to exposure led to a significant reduction in damage [126]. Similarly, a coupled water-energy model for the Laotian-Thai grid was developed with a focus on the Mekong River basin, investigating the risk posed by prolonged droughts on hydropower production, electricity costs, and carbon dioxide emissions [127]. Climate risks have also been represented as correlated hydrometeorological processes that influence both supply and demand in electricity markets throughout the western United States [32].

Increasing simulation lengths and ensemble sizes by several orders of magnitude to credibly assess climate risks poses a significant computational challenge. To address this computational bottleneck, many studies have explored the use of emulators, idealized models, hierarchical models, and surrogate models to model and mitigate climate risks in infrastructure systems [56, 128–132].

Adaptive and Robust Adaptation Pathways

The unique challenges in assessing and managing climate risks call for the integration of multidisciplinary insights, the incorporation of ethical values into decision-making, and the rigorous characterization and quantification of uncertainties and their impacts on metrics relevant to stakeholders [96, 133]. Bottom-up methodologies are particularly illuminating in this context. Eschewing the traditional "predict-then-plan" approach, these methodologies embrace a comprehensive exploration of plausible scenarios, assessing potential system responses before evaluating the likelihood of each scenario. Exploratory modeling, for example, improves understanding of potential system behaviors, facilitates hypothesis formulation, and helps focus attention on scenarios that

merit in-depth analysis [134]. Methods like decision scaling and robust decision making also emphasize exploring interactions between different decisions and possible futures to identify robust plans [135–139]. Moreover, recognizing infrastructure planning as a dynamic, sequential problem, methodologies such as Real Options Analysis, Reinforcement Learning, and Direct Policy Search highlight the importance of adaptability and flexibility for efficacious, robust long-term planning [140–146].

While these approaches are less common in long-term electricity planning systems, they are valuable for identifying decisions that meet multiple objectives, such as minimizing cost, improving resilience, decarbonizing, and protecting nature, under deep uncertainty. Several promising approaches may support these goals. One such approach is optimizing explicitly for robustness. Alternatively, employing subjective probabilities over possible futures allows exploration of low-probability scenarios without biasing final results [107]. Third, where robustness is not considered in the optimization step, stress-testing plausible alternatives can be an insightful exploratory step. These methods, when employed judiciously, pave the way for long-term electricity planning that adeptly addresses the complexities and uncertainties of climate risks, fostering the development of resilient and adaptive electricity systems.

Summary

This paper critically examines the multifaceted dimensions of climate risk assessment and management, with a particular emphasis on electricity systems. Climate change poses significant risks to electricity systems, with its direct and cascading effects causing severe consequences. Additionally, emerging technologies are transforming energy demand and load profiles, potentially adding additional stress to the grid. It is evident that climate risks are already impacting electricity systems and the stresses are likely to escalate in the future due to the changing climate patterns and technology trends.

The studies considered represent climate risks using short weather records and a small number of technological and economic scenarios, which limits the extent to which climate risks to the modeled electricity systems can be quantified. There is thus a pressing need to improve the representation of extreme weather events and long-term climate change, embrace multisector dynamics, and critically evaluate uncertainties in long-term electricity systems planning studies. This will require adopting integrated modeling, broadening scenario analyses beyond standardized scenarios, developing use-tailored weather generators, and employing adaptive and robust methodologies that account for system dynamics and uncertainties.

Acknowledgements The authors thank Vivek Srikrishnan and David J. Farnham for insightful comments and suggestions on preliminary versions of this manuscript.

Author Contributions JDG designed the study and drafted the manuscript. JDG, YA, KS, and DC contributed to the literature review and manuscript revisions.

Funding The authors declare that no funds, grants, or other support was received during the preparation of this manuscript.

Declarations

Conflict of Interest The authors declare no competing interests.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

References

1. Jain Y, Jain R. India and Pakistan emerge as early victims of extreme heat conditions due to climate injustice. BMJ. 2022;377:1207. https://doi.org/10.1136/bmj.o1207. (**Chap Opinion**).
2. Doss-Gollin J, Farnham DJ, Lall U, Modi V, How unprecedented was the February 2021 Texas cold snap? Environ Res Lett; 2021. https://doi.org/10.1088/1748-9326/ac0278.
3. Busby JW, Baker K, Bazilian MD, Gilbert AQ, Grubert E, Rai V, Rhodes JD, Shidore S, Smith CA, Webber ME. Cascading risks: understanding the 2021 winter blackout in Texas. Energy Res Soc Sci. 2021;77. https://doi.org/10.1016/j.erss.2021.102106.
4. Shahid S. Vulnerability of the power sector of Bangladesh to climate change and extreme weather events. Reg Environ Change. 2012;12(3):595–606. https://doi.org/10.1007/s10113-011-0276-z.
5. Kwasinski A, Andrade F, Castro-Sitiriche MJ, O'Neill-Carrillo E. Hurricane Maria effects on Puerto Rico electric power infrastructure. IEEE Power Energy Technol Syst J. 2019;6(1):85–94. https://doi.org/10.1109/JPETS.2019.2900293.
6. Do V, McBrien H, Flores NM, Northrop AJ, Schlegelmilch J, Kiang MV, Casey JA. Spatiotemporal distribution of power outages with climate events and social vulnerability in the USA. Nat Commun. 2023;14(1):2470. https://doi.org/10.1038/s41467-023-38084-6.
7. Stone BJ, Gronlund CJ, Mallen E, Hondula D, O'Neill MS, Rajput M, Grijalva S, Lanza K, Harlan S, Larsen L, Augenbroe G, Krayenhoff ES, Broadbent A, Georgescu M. How blackouts during heat waves amplify mortality and morbidity risk. Environ Sci Technol. 2023. https://doi.org/10.1021/acs.est.2c09588.
8. International Energy Agency. World Energy Outlook 2022. International Energy Agency: Technical report; 2022.
9. Meinshausen M, Lewis J, McGlade C, Gütschow J, Nicholls Z, Burdon R, Cozzi L, Hackmann B. Realization of Paris Agreement pledges may limit warming just below 2 °C. Nature. 2022;604(7905):304–9. https://doi.org/10.1038/s41586-022-04553-z.
10. Auffhammer M, Baylis P, Hausman CH. Climate change is projected to have severe impacts on the frequency and intensity of peak electricity demand across the United States. Proc Natl

Acad Sci. 2017;114(8):1886–91. https://doi.org/10.1073/pnas.1613193114.
11. Romitti Y, Sue Wing I. Heterogeneous climate change impacts on electricity demand in world cities circa mid-century. Sci Rep. 2022;12(1):4280. https://doi.org/10.1038/s41598-022-07922-w.
12. Amonkar YV, Doss-Gollin J, Farnham DJ, Lall U, Modi V. Differential effects of climate change on average and peak demand for heating and cooling across the contiguous United States. Earth ArXiv. 2023. https://doi.org/10.31223/X5TH5X.
13. Plaga LS, Bertsch V. Methods for assessing climate uncertainty in energy system models-a systematic literature review. Appl Energy. 2023;331. https://doi.org/10.1016/j.apenergy.2022.120384.
14. Jung C, Schindler D. A review of recent studies on wind resource projections under climate change. Renew Sust Energ Rev. 2022;165: 112596. https://doi.org/10.1016/j.rser.2022.112596.
15. Koliou M, van de Lindt JW, McAllister TP, Ellingwood BR, Dillard M, Cutler H. State of the research in community resilience: progress and challenges. Sustain Resilient Infr. 2020;5(3):131–51. https://doi.org/10.1080/23789689.2017.1418547.
16. Yalew SG, van Vliet MTH, Gernaat DEHJ, Ludwig F, Miara A, Park C, Byers E, De Cian E, Piontek F, Iyer G, Mouratiadou I, Glynn J, Hejazi M, Dessens O, Rochedo P, Pietzcker R, Schaeffer R, Fujimori S, Dasgupta S, Mima S, da Silva SRS, Chaturvedi V, Vautard R, van Vuuren DP. Impacts of climate change on energy systems in global and regional scenarios. Nat Energy. 2020;5(10):794–802. https://doi.org/10.1038/s41560-020-0664-z.
17. Schaeffer R, Szklo AS, Pereira de Lucena AF, Moreira Cesar Borba BS, Pupo Nogueira LP, Fleming FP, Troccoli A, Harrison M, Boulahya MS. Energy sector vulnerability to climate change: a review. Energy 2012;38(1)1–12. https://doi.org/10.1016/j.energy.2011.11.056.
18. Clarke L, Wei Y-M, Navarro ADLV, Garg A, Hahmann AN, Khennas S, Azevedo IML, Löschel A, Singh AK, Steg L, Strbac G, Wada K. Energy systems. In: Shukla PR, Skea J, Slade R, Khourdajie AA, van Diemen R, McCollum D, Pathak M, Some S, Vyas P, Fradera R, Belkacemi M, Hasija A, Lisboa G, Luz S, Malley J (eds) Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, UK and New York, NY, USA (2022). Chap. 6. https://doi.org/10.1017/9781009157926.008.
19. Maia-Silva D, Kumar R, Nateghi R. The critical role of humidity in modeling summer electricity demand across the United States. Nat Commun. 2020;11(1):1686. https://doi.org/10.1038/s41467-020-15393-8.
20. Henley A, Peirson J. Non-linearities in electricity demand and temperature: parametric versus non-parametric methods. Oxf Bull Econ Stat. 1997;59(1):149–62. https://doi.org/10.1111/1468-0084.00054.
21. Moral-Carcedo J, Vicéns-Otero J. Modelling the non-linear response of Spanish electricity demand to temperature variations. Energy Econ. 2005;27(3):477–94. https://doi.org/10.1016/j.eneco.2005.01.003.
22. Lee J, Dessler AE. The impact of neglecting climate change and variability on ERCOT's forecasts of electricity demand in Texas. Weather Clim Soc. 2022;14(2):499–505. https://doi.org/10.1175/WCAS-D-21-0140.1. (**Chap. Weather, Climate, and Society**).
23. Shaffer, B., Quintero, D., Rhodes, J.: Changing sensitivity to cold weather in Texas power demand. iScience. 2022;25(4), 104173. https://doi.org/10.1016/j.isci.2022.104173
24. Alipour P, Mukherjee S, Nateghi R. Assessing climate sensitivity of peak electricity load for resilient power systems planning and operation: a study applied to the Texas region. Energy. 2019;185:1143–53. https://doi.org/10.1016/j.energy.2019.07.074.
25. Bett PE, Thornton HE. The climatological relationships between wind and solar energy supply in Britain. Renew Energy. 2016;87:96–110. https://doi.org/10.1016/j.renene.2015.10.006.
26. Craig MT, Jaramillo P, Hodge B-M, Nijssen B, Brancucci C. Compounding climate change impacts during high stress periods for a high wind and solar power system in Texas. Environ Res Lett. 2020;15(2): 024002. https://doi.org/10.1088/1748-9326/ab6615.
27. Jerez S, Tobin I, Vautard R, Montávez JP, López-Romero JM, Thais F, Bartok B, Christensen OB, Colette A, Déqué M, Nikulin G, Kotlarski S, van Meijgaard E, Teichmann C, Wild M. The impact of climate change on photovoltaic power generation in Europe. Nat Commun. 2015;6(1):10014. https://doi.org/10.1038/ncomms10014.
28. Lundquist JK, DuVivier KK, Kaffine D, Tomaszewski JM. Costs and consequences of wind turbine wake effects arising from uncoordinated wind energy development. Nat Energy. 2019;4(1):26–34. https://doi.org/10.1038/s41560-018-0281-2.
29. van Vliet MTH, van Beek LPH, Eisner S, Flörke M, Wada Y, Bierkens MFP. Multi-model assessment of global hydropower and cooling water discharge potential under climate change. Glob Environ Change. 2016;40:156–70. https://doi.org/10.1016/j.gloenvcha.2016.07.007.
30. van Vliet MTH, Yearsley JR, Ludwig F, Vögele S, Lettenmaier DP, Kabat P. Vulnerability of US and European electricity supply to climate change. Nature Climate Change. 2012;2(9):676–81. https://doi.org/10.1038/nclimate1546.
31. Kern JD, Su Y, Hill J. A retrospective study of the 2012–2016 California drought and its impacts on the power sector. Environ Res Lett. 2020;15(9): 094008. https://doi.org/10.1088/1748-9326/ab9db1.
32. Su Y, Kern JD, Denaro S, Hill J, Reed P, Sun Y, Cohen J, Characklis GW. An open source model for quantifying risks in bulk electric power systems from spatially and temporally correlated hydrometeorological processes. Environ Modell Softw. 2020;126: 104667. https://doi.org/10.1016/j.envsoft.2020.104667.
33. Loew A, Jaramillo P, Zhai H, Ali R, Nijssen B, Cheng Y, Klima K. Fossil fuel-fired power plant operations under a changing climate. Clim Change. 2020;163(1):619–32. https://doi.org/10.1007/s10584-020-02834-y.
34. Coffel ED, Mankin JS. Thermal power generation is disadvantaged in a warming world. Environ Res Lett. 2020. https://doi.org/10.1088/1748-9326/abd4a8.
35. Liu L, Hejazi M, Li H, Forman B, Zhang X. Vulnerability of US thermoelectric power generation to climate change when incorporating state-level environmental regulations. Nat Energy. 2017;2(8):1–5. https://doi.org/10.1038/nenergy.2017.109.
36. Otero N, Martius O, Allen S, Bloomfield H, Schaefli B. Characterizing renewable energy compound events across Europe using a logistic regression-based approach. Meteorol Appl. 2022; 29(5). https://doi.org/10.1002/met.2089
37. Li B, Basu S, Watson SJ, Russchenberg HWJ. A brief climatology of dunkelflaute events over and surrounding the North and Baltic Sea areas. Energies. 2021;14(20). https://doi.org/10.3390/en14206508.
38. Boston A, Bongers GD, Bongers N. Characterisation and mitigation of renewable droughts in the Australian National Electricity Market. Environ Res Commun. 2022;4(3): 031001. https://doi.org/10.1088/2515-7620/ac5677.
39. Ohba M, Kanno Y, Bando S. Effects of meteorological and climatological factors on extremely high residual load and possible future changes. Renew Sustain Energy Rev. 2023;175: 113188. https://doi.org/10.1016/j.rser.2023.113188.
40. Brown PT, Farnham DJ, Caldeira K. Meteorology and climatology of historical weekly wind and solar power resource droughts over

western North America in ERA5. SN Appl Sci. 2021;3(10):814. https://doi.org/10.1007/s42452-021-04794-z.
41. Ghil M, Lucarini V. The physics of climate variability and climate change. Rev Modern Phys. 2020;92(3): 035002. https://doi.org/10.1103/revmodphys.92.035002.
42. Doss-Gollin J, Farnham DJ, Steinschneider S, Lall U. Robust adaptation to multiscale climate variability. Earths Future. 2019;7(7):734–47. https://doi.org/10.1029/2019ef001154.
43. Ely CR, Brayshaw DJ, Methven J, Cox J, Pearce O. Implications of the North Atlantic Oscillation for a UK-Norway Renewable power system. Energy Policy. 2013;62:1420–7. https://doi.org/10.1016/j.enpol.2013.06.037.
44. Jerez S, Trigo RM. Time-scale and extent at which large-scale circulation modes determine the wind and solar potential in the Iberian Peninsula. Environ Res Lett. 2013;8(4): 044035. https://doi.org/10.1088/1748-9326/8/4/044035.
45. Neubacher C, Witthaut D, Wohland J. Multi-decadal offshore wind power variability can be mitigated through optimized European allocation. Adv Geosci. 2021;54:205–15. https://doi.org/10.5194/adgeo-54-205-2021.
46. Bartos M, Chester M, Johnson N, Gorman B, Eisenberg D, Linkov I, Bates M. Impacts of rising air temperatures on electric transmission ampacity and peak electricity load in the United States. Environ Res Lett. 2016;11(11): 114008. https://doi.org/10.1088/1748-9326/11/11/114008.
47. Feng X, Yang J, Luo C, Sun Y, Liu M, Tang Y. A risk evaluation method for cascading failure considering transmission line icing. In: 2015 IEEE Innovative Smart Grid Technologies - Asia (ISGT ASIA). 2015; pp. 1–4. https://doi.org/10.1109/ISGT-Asia.2015.7387014.
48. Yaji K, Homma H, Sakata G, Watanabe M. Evaluation on flashover voltage property of snow accreted insulators for overhead transmission lines, part I - field observations and laboratory tests to evaluate snow accretion properties. IEEE Trans Dielectr Electr Insul. 2014;21(6):2549–58. https://doi.org/10.1109/TDEI.2014.004564.
49. Croce P, Formichi P, Landi F, Mercogliano P, Bucchignani E, Dosio A, Dimova S. The snow load in Europe and the climate change. Clim Risk Manag. 2018;20:138–54. https://doi.org/10.1016/j.crm.2018.03.001.
50. Dian S, Cheng P, Ye Q, Wu J, Luo R, Wang C, Hui D, Zhou N, Zou D, Yu Q, Gong X. Integrating wildfires propagation prediction into early warning of electrical transmission line outages. IEEE Access. 2019;7:27586–603. https://doi.org/10.1109/ACCESS.2019.2894141.
51. Ranasinghe R, Ruane AC, Vautard R, Arnell N, Coppola E, Cruz FA, Dessai S, Islam AS, Rahimi M, Ruiz Carrascal D, Sillmann J, Sylla MB, Tebaldi C, Wang W, Zaaboul R. Climate change information for regional impact and for risk assessment. In: Masson-Delmotte V, Zhai P, Pirani A, Connors SL, Péan C, Berger S, Caud N, Chen Y, Goldfarb L, Gomis MI, Huang M, Leitzell K, Lonnoy E, Matthews JBR, Maycock TK, Waterfield T, Yelekçi O, Yu R, Zhou B (eds.) Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press,(2021). Chap. 12. https://doi.org/10.1017/9781009157896.014.
52. Committee on enhancing the resilience of the nation's electric power transmission and distribution system, Board on Energy and Environmental Systems, Division on Engineering and Physical Sciences, National Academies of Sciences, Engineering, and Medicine: Enhancing the Resilience of the Nation's Electricity System, p. 24836. Washington, D.C: National Academies Press, 2017. https://doi.org/10.17226/24836
53. Seneviratne SI, Zhang X, Adnan M, Badi W, Dereczynski C, Di Luca A, Ghosh S, Iskandar I, Kossin J, Lewis S, Otto F, Pinto I, Satoh M, Vicente-Serrano SM, Wehner M, Zhou B. Weather and climate extreme events in a changing climate. In: Masson-Delmotte V, Zhai P, Pirani A, Connors SL, Péan C, Berger S, Caud N, Chen Y, Goldfarb L, Gomis MI, Huang M, Leitzell K, Lonnoy E, Matthews JBR, Maycock TK, Waterfield T, Yelekçi O, Yu R, Zhou B (eds) Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press, 2021. Chap. 11. https://doi.org/10.1017/9781009157896.013.
54. Knutson T, Camargo SJ, Chan JCL, Emanuel K, Ho C-H, Kossin J, Mohapatra M, Satoh M, Sugi M, Walsh K, Wu L. Tropical cyclones and climate change assessment: part II: projected response to anthropogenic warming. Bull Am Meteorol Soc. 2020;101(3):303–22. https://doi.org/10.1175/BAMS-D-18-0194.1. (**Chap. Bulletin of the American Meteorological Society**).
55. Sobel AH, Wing AA, Camargo SJ, Patricola CM, Vecchi GA, Lee C-Y, Tippett MK. Tropical cyclone frequency. Earths. Future. 2021;9(12):2021–002275. https://doi.org/10.1029/2021EF002275.
56. Kopp RE, DeConto RM, Bader DA, Hay CC, Horton RM, Kulp S, Oppenheimer M, Pollard D, Strauss BH. Evolving understanding of Antarctic ice-sheet physics and ambiguity in probabilistic sea-level projections. Earths Fut. 2017;5(12):1217–33. https://doi.org/10.1002/2017ef000663.
57. Gori A, Lin N, Xi D, Emanuel K. Tropical cyclone climatology change greatly exacerbates US extreme rainfall-surge hazard. Nat Clim Change. 2022;12(2):171–8. https://doi.org/10.1038/s41558-021-01272-7.
58. Flannigan M, Cantin AS, de Groot WJ, Wotton M, Newbery A, Gowman LM. Global wildland fire season severity in the 21st century. For Ecol Manag. 2013;294:54–61. https://doi.org/10.1016/j.foreco.2012.10.022.
59. Lee J-Y, Marotzke J, Bala G, Cao L, Corti S, Dunne JP, Engelbrecht F, Fischer E, Fyfe JC, Jones C, Maycock A, Mutemi J, Ndiaye O, Panickal S, Zhou T. Future global climate: scenario-based projections and near-term information. In: Masson-Delmotte V, Zhai P, Pirani A, Connors SL, Péan C, Berger S, Caud N, Chen Y, Goldfarb L, Gomis MI, Huang M, Leitzell K, Lonnoy E, Matthews JBR, Maycock TK, Waterfield T, Yelekçi O, Yu R, Zhou B (eds) Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press; 2021. Chap. 4. https://doi.org/10.1017/9781009157896.006.
60. Santer BD, Po-Chedley S, Zelinka MD, Cvijanovic I, Bonfils C, Durack PJ, Fu Q, Kiehl J, Mears C, Painter J, Pallotta G, Solomon S, Wentz FJ, Zou C-Z. Human influence on the seasonal cycle of tropospheric temperature. Science. 2018;361(6399). https://doi.org/10.1126/science.aas8806.
61. Donohoe A, Battisti DS. The seasonal cycle of atmospheric heating and temperature. J Clim. 2013;26(14):4962–80. https://doi.org/10.1175/JCLI-D-12-00713.1. (**Chap. Journal of Climate**).
62. Shaw TA, Baldwin M, Barnes EA, Caballero R, Garfinkel CI, Hwang YT, Li C, O'Gorman PA, Rivière G, Simpson IR, Voigt A. Storm track processes and the opposing influences of climate change. Nat Geosci. 2016;9(9):656–64. https://doi.org/10.1038/ngeo2783.
63. Nabizadeh E, Hassanzadeh P, Yang D, Barnes EA. Size of the atmospheric blocking events: scaling law and response to climate change. Geophys Res Lett. 2019;46(22):13488–99. https://doi.org/10.1029/2019gl084863.

64. Gulev SK, Thorne PW, Ahn J, Dentener FJ, Domingues CM, Gerland S, Gong D, Kaufman DS, Nnamchi HC, Quaas J, Rivera JA, Sathyendranath S, Smith SL, Trewin B, von Schuckmann K, Vose RS. Changing state of the climate system. In: Masson-Delmotte V, Zhai P, Pirani A, Connors S.L, Péan C, Berger S, Caud N, Chen Y, Goldfarb L, Gomis MI, Huang M, Leitzell K, Lonnoy E, Matthews JBR, Maycock TK, Waterfield T, Yelekçi, O, Yu R, Zhou B (eds) Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press, 2021;Chap. 2. https://doi.org/10.1017/9781009157896.004.
65. Petoukhov V, Petri S, Rahmstorf S, Coumou D, Kornhuber K, Schellnhuber HJ. Role of quasiresonant planetary wave dynamics in recent boreal spring-to-autumn extreme events. Proc Nat Acad Sci Unit Am. 2016;113(25):6862–7. https://doi.org/10.1073/pnas.1606300113.
66. Kornhuber K, Osprey S, Coumou D, Petri S, Petoukhov V, Rahmstorf S, Gray L. Extreme weather events in early summer 2018 connected by a recurrent hemispheric wave-7 pattern. Environ Res Lett. 2019;14(5): 054002. https://doi.org/10.1088/1748-9326/ab13bf.
67. Trenberth KE, Fasullo JT. Climate extremes and climate change: the Russian heat wave and other climate extremes of 2010. J Geophys Res Atmos. 2012;117:D17. https://doi.org/10.1029/2012jd018020.
68. Pfleiderer P, Schleussner C-F, Kornhuber K, Coumou D. Summer weather becomes more persistent in a 2 °C world. Nat Clim Change. 2019;1–6. https://doi.org/10.1038/s41558-019-0555-0.
69. Miller NL, Hayhoe K, Jin J, Auffhammer M. Climate, extreme heat, and electricity demand in California. J Appl Meteorol Climatol. 2008;47(6):1834–44. https://doi.org/10.1175/2007JAMC1480.1. (**Chap. Journal of Applied Meteorology and Climatology**).
70. Cohen J, Zhang X, Francis J, Jung T, Kwok R, Overland J, Ballinger TJ, Bhatt US, Chen HW, Coumou D, Feldstein S, Gu H, Handorf D, Henderson G, Ionita M, Kretschmer M, Laliberte F, Lee S, Linderholm HW, Maslowski W, Peings Y, Pfeiffer K, Rigor I, Semmler T, Stroeve J, Taylor PC, Vavrus S, Vihma T, Wang S, Wendisch M, Wu Y, Yoon J. Divergent consensuses on Arctic amplification influence on midlatitude severe winter weather. Nat Clim Change. 2020;10(1):20–9. https://doi.org/10.1038/s41558-019-0662-y.
71. Tobin I, Greuell W, Jerez S, Ludwig F, Vautard R, van Vliet MTH, Bréon F-M. Vulnerabilities and resilience of European power generation to 1.5 °C, 2 °C and 3 °C warming. Environ Res Lett. 2018;13(4),044024. https://doi.org/10.1088/1748-9326/aab211.
72. Bonanno R, Viterbo F, Maurizio RG. Climate change impacts on wind power generation for the Italian peninsula. Reg Environ Change. 2023;23(1). https://doi.org/10.1007/s10113-022-02007-w.
73. Hahmann AN, García-Santiago O, Peña A. Current and future wind energy resources in the North Sea according to CMIP6. Wind Energy Sci. 2022;7(6):2373–91. https://doi.org/10.5194/wes-7-2373-2022.
74. Wohland J. Process-based climate change assessment for European winds using EURO-CORDEX and global models. Environ Res Lett. 2022;17(12). https://doi.org/10.1088/1748-9326/aca77f.
75. Gonzalez PLM, Brayshaw DJ, Zappa G. The contribution of North Atlantic atmospheric circulation shifts to future wind speed projections for wind power over Europe. Clim Dyn. 2019;53(7–8):4095–113. https://doi.org/10.1007/s00382-019-04776-3.
76. Wasti A, Ray P, Wi S, Folch C, Ubierna M, Karki P. Climate change and the hydropower sector: a global review. WIREs Clim Change. 2022;13(2):757. https://doi.org/10.1002/wcc.757.
77. Dodman D, Hayward B, Pelling M, Castan Broto V, Chow W, Chu E, Dawson R, Khirfan L, McPhearson T, Prakash A, Zheng Y, Ziervogel G. Cities, settlements and key infrastructure. In: Pörtner HO, Roberts DC, Tignor M, Poloczanska ES, Mintenbeck K, Alegría A, Craig M, Langsdorf S, Löschke S, Möller V, Okem A, Rama B (eds) Climate change 2022: impacts, adaptation and vulnerability. Contribution ofWorking Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press, 2022;Chap. 6. https://doi.org/10.1017/9781009325844.008.
78. Carvajal PE, Li FGN, Soria R, Cronin J, Anandarajah G, Mulugetta Y. Large hydropower, decarbonisation and climate change uncertainty: modelling power sector pathways for Ecuador. Energy Strat Rev. 2019;23:86–99. https://doi.org/10.1016/j.esr.2018.12.008.
79. Dittes B, Špačková O, Schoppa L, Straub D. Managing uncertainty in flood protection planning with climate projections. Hydrol Earth Syst Sci. 2018;22(4):2511–26. https://doi.org/10.5194/hess-22-2511-2018.
80. Lafferty DC, Sriver RL. Downscaling and bias-correction contribute considerable uncertainty to local climate projections in CMIP6. Preprints. 2023. https://doi.org/10.22541/essoar.168286894.44910061/v1.
81. Perera ATD, Nik VM, Chen D, Scartezzini J-L, Hong T. Quantifying the impacts of climate change and extreme climate events on energy systems. Nat Energy. 2020;5(2):150–9. https://doi.org/10.1038/s41560-020-0558-0.
82. DeAngelo J, Azevedo I, Bistline J, Clarke L, Luderer G, Byers E, Davis SJ. Energy systems in scenarios at net-zero CO2 emissions. Nat Commun. 2021;12(1):6096. https://doi.org/10.1038/s41467-021-26356-y.
83. Kondi-Akara G, Hingray B, Francois B, Diedhiou A. Recent trends in urban electricity consumption for cooling in West and Central African countries. Energy. 2023;276: 127597. https://doi.org/10.1016/j.energy.2023.127597.
84. Waite M, Modi V. Electricity load implications of space heating decarbonization pathways. Joule. 2020;4(2):376–94. https://doi.org/10.1016/j.joule.2019.11.011.
85. Hausfather Z, Peters GP. Emissions-the 'business as usual' story is misleading. Nature. 2020;577(7792):618–20. https://doi.org/10.1038/d41586-020-00177-3.
86. Walker WE, Lempert RJ, Kwakkel JH. Deep uncertainty. In: Gass SI, Fu MC (eds) Encyclopedia of operations research and management science. pp. 395–402. Boston, MA: Springer US, 2013. https://doi.org/10.1007/978-1-4419-1153-7_1140.
87. Mathy S, Criqui P, Knoop K, Fischedick M, Samadi S. Uncertainty management and the dynamic adjustment of deep decarbonization pathways. Clim Policy. 2016;16(sup1):47–62. https://doi.org/10.1080/14693062.2016.1179618.
88. Sepulveda NA, Jenkins JD, Edington A, Mallapragada DS, Lester RK. The design space for long-duration energy storage in decarbonized power systems. Nat Energy. 2021;6(5):506–16. https://doi.org/10.1038/s41560-021-00796-8.
89. Larson E, Greig C, Jenkins J, Mayfield E, Pascale A, Zhang C, Drossman J, Williams R, Pacala S, Socolow R, Baik E, Birdsey R, Duke R, Jones R, Haley B, Leslie E, Paustian K, Swan A. Net-zero America: potential pathways, infrastructure, and impacts. Technical report. Princeton: Princeton University; 2021.
90. Denholm P, Brown P, Cole W, Mai T, Sergi B. Examining supply-side options to achieve 100% clean electricity by 2035. Technical Report NREL/TP-6A40-81644. Golden: National Renewable Energy Laboratory, 2022. https://doi.org/10.2172/1885591
91. Walter L, Jantarasami L, Schneider C. Pathways to net-zero emissions. Decarb America: Technical report; 2021.

92. Shaner MR, Davis SJ, Lewis NS, Caldeira K. Geophysical constraints on the reliability of solar and wind power in the United States. Energy Environ Sci. 2018;11(4):914–25. https://doi.org/10.1039/C7EE03029K.
93. Tong D, Farnham DJ, Duan L, Zhang Q, Lewis NS, Caldeira K, Davis SJ. Geophysical constraints on the reliability of solar and wind power worldwide. Nat Communications. 2021;12(1):6146. https://doi.org/10.1038/s41467-021-26355-z.
94. The Nature Conservancy. Power of place: national. The Nature Conservancy: Technical report; 2023.
95. Pianosi F, Beven K, Freer J, Hall JW, Rougier J, Stephenson DB, Wagener T. Sensitivity analysis of environmental models: a systematic review with practical workflow. Environ Modell Softw. 2016;79:214–32. https://doi.org/10/f8n6zw
96. Srikrishnan V, Lafferty DC, Wong TE, Lamontagne JR, Quinn JD, Sharma S, Molla NJ, Herman JD, Sriver RL, Morris JF, Lee BS. Uncertainty analysis in multi-sector systems: considerations for risk analysis, projection, and planning for complex systems. Earths Fut. 2022;10(8):2021–002644. https://doi.org/10.1029/2021EF002644.
97. Saltelli A, Ratto M, Andres T, Campolongo F, Cariboni J, Gatelli D, Saisana M, Tarantola S. Global sensitivity analysis: the primer. John Wiley & Sons, Ltd, 2008.
98. Stephens GL, L'Ecuyer T, Forbes R, Gettlemen A, Golaz J-C, Bodas-Salcedo A, Suzuki K, Gabriel P, Haynes J. Dreary state of precipitation in global models. J Geophys Res Atmosp. 2010;115(D24):24211. https://doi.org/10.1029/2010jd014532.
99. Muller CJ, O'Gorman PA, Back LE. Intensification of precipitation extremes with warming in a cloud-resolving model. J Clim. 2011;24(11):2784–800. https://doi.org/10.1175/2011jcli3876.1.
100. Espinoza V, Waliser DE, Guan B, Lavers DA, Ralph FM. Global analysis of climate change projection effects on atmospheric rivers. Geophys Res Lett. 2018;47(3):514. https://doi.org/10.1029/2017gl076968.
101. Smith DM, Scaife AA, Eade R, Athanasiadis P, Bellucci A, Bethke I, Bilbao R, Borchert LF, Caron L-P, Counillon F, Danabasoglu G, Delworth T, Doblas-Reyes FJ, Dunstone NJ, Estella-Perez V, Flavoni S, Hermanson L, Keenlyside N, Kharin V, Kimoto M, Merryfield WJ, Mignot J, Mochizuki T, Modali K, Monerie P-A, Müller WA, Nicolí D, Ortega P, Pankatz K, Pohlmann H, Robson J, Ruggieri P, Sospedra-Alfonso R, Swingedouw D, Wang Y, Wild S, Yeager S, Yang X, Zhang L. North Atlantic climate far more predictable than models imply. Nature. 2020;583(7818):796–800. https://doi.org/10.1038/s41586-020-2525-0.
102. Kravtsov S. Pronounced differences between observed and CMIP5-simulated multidecadal climate variability in the twentieth century. Geophys Res Lett. 2017;44(11):5749–57. https://doi.org/10.1002/2017GL074016.
103. Greene AM, Robertson AW. Interannual and low-frequency variability of Upper Indus Basin winter/spring precipitation in observations and CMIP5 models. Clim Dynam. 2017;49(11–12):4171–88. https://doi.org/10.1007/s00382-017-3571-7.
104. Feng J, Lian T, Ying J, Li J, Li G. Do CMIP5 models show El Niño diversity? J Clim. 2019. https://doi.org/10.1175/jcli-d-18-0854.1.
105. Srikrishnan V, Guan Y, Tol RSJ, Keller K. Probabilistic projections of baseline twenty-first century CO2 emissions using a simple calibrated integrated assessment model. Climatic Change. 2022;170(3):37. https://doi.org/10.1007/s10584-021-03279-7.
106. Milly PCD, Betancourt J, Falkenmark M, Hirsch RM, Kundzewicz ZW, Lettenmaier DP, Stouffer RJ. Stationarity is dead: whither water management? Science. 2008;319(5863):573–4. https://doi.org/10.1126/science.1151915.
107. Doss-Gollin J, Keller K. A subjective Bayesian framework for synthesizing deep uncertainties in climate risk management. Earths Fut. 2023; 11(1). https://doi.org/10.1029/2022EF003044.
108. Jain S, Lall U. Floods in a changing climate: does the past represent the future? Water Resour Res. 2001;37(12):3193–205. https://doi.org/10.1029/2001wr000495.
109. Dowling JA, Rinaldi KZ, Ruggles TH, Davis SJ, Yuan M, Tong F, Lewis NS, Caldeira K. Role of long-duration energy storage in variable renewable electricity systems. Joule. 2020;4(9):1907–28. https://doi.org/10.1016/j.joule.2020.07.007.
110. Collins S, Deane P, Ó Gallachóir B, Pfenninger S, Staffell I. Impacts of inter-annual wind and solar variations on the European power system. Joule. 2018; 2(10), 2076–2090. https://doi.org/10.1016/j.joule.2018.06.020.
111. Lanzante JR, Dixon KW, Nath MJ, Whitlock CE, Adams-Smith D. Some pitfalls in statistical downscaling of future climate. Bull Am Meteorol Soc. 2018;99(4):791–803. https://doi.org/10.1175/bams-d-17-0046.1.
112. Ehret U, Zehe E, Wulfmeyer V, Warrach-Sagi K, Liebert J. Should we apply bias correction to global and regional climate model data? Hydrol Earth Syst Sci. 2012;16(9):3391–404. https://doi.org/10.5194/hess-16-3391-2012.
113. Harder P, Yang Q, Ramesh V, Hernandez-Garcia A, Watson CD, Szwarcman D, Rolnick D. Generating physically-consistent high-resolution climate data with hard-constrained neural networks. In: AAAI 2022 Fall Symposium: The Role of AI in Responding To Climate Challenges. 2022, p 7.
114. Price I, Rasp S. Increasing the accuracy and resolution of precipitation forecasts using deep generative models. arXiv. 2022. https://doi.org/10.48550/arXiv.2203.12297.
115. Lall U, Sharma A. A nearest neighbor bootstrap for resampling hydrologic time series. Water Resources Research. 1996;32(3):679–93. https://doi.org/10.1029/95WR02966.
116. Rajagopalan B, Lall U. A k-nearest-neighbor simulator for daily precipitation and other weather variables. Water Resour Res. 1999;35(10):3089–101. https://doi.org/10.1029/1999WR900028.
117. Amonkar Y, Farnham DJ, Lall U. A k-nearest neighbor space-time simulator with applications to large-scale wind and solar power modeling. Patterns. 2022;3(3): 100454. https://doi.org/10.1016/j.patter.2022.100454.
118. Papalexiou SM, Serinaldi F, Clark MP. Large-domain multisite precipitation generation: operational blueprint and demonstration for 1,000 sites. Water Resour Res. 2023;59(3):2022–034094. https://doi.org/10.1029/2022WR034094.
119. Steinschneider S, Brown C. A semiparametric multivariate, multisite weather generator with low-frequency variability for use in climate risk assessments. Water Resour Res. 2013;49(11):7205–20. https://doi.org/10.1002/wrcr.20528.
120. Gupta RS, Steinschneider S, Reed PM. Understanding contributions of Paleo-informed natural variability and climate changes on hydroclimate extremes in the Central Valley region of California. Preprint, Preprints, 2023. https://doi.org/10.22541/essoar.167870424.46495295/v1.
121. Bracken C, Rajagopalan B, Zagona E. A hidden Markov model combined with climate indices for multidecadal streamflow simulation. Water Resour Res. 2014;50(10):7836–46. https://doi.org/10.1002/2014WR015567.
122. ...Merz B, Aerts JCJH, Arnbjerg-Nielsen K, Baldi M, Becker A, Bichet A, Blöschl G, Bouwer LM, Brauer A, Cioffi F, Delgado JM, Gocht M, Guzzetti F, Harrigan S, Hirschboeck K, Kilsby C, Kron W, Kwon HH, Lall U, Merz R, Nissen K, Salvatti P, Swierczynski T, Ulbrich U, Viglione A, Ward PJ, Weiler M, Wilhelm B, Nied M. Floods and climate: emerging perspectives for flood risk assessment and management. Nat Hazards Earth Syst Sci. 2014;14(7):1921–42. https://doi.org/10/gb9nzm
123. Lamontagne JR, Reed PM, Link R, Calvin KV, Clarke LE, Edmonds JA. Large ensemble analytic framework for

consequence-driven discovery of climate change scenarios. Earths Fut. 2018;6(3):488–504. https://doi.org/10.1002/2017ef000701.
124. Dolan F, Lamontagne J, Link R, Hejazi M, Reed P, Edmonds J. Evaluating the economic impact of water scarcity in a changing world. Nat Commun. 2021;12(1):1915.
125. Zarekarizi M, Srikrishnan V, Keller K. Neglecting uncertainties biases house-elevation decisions to manage riverine flood risks. Nat Commun. 2020;11(1):5361. https://doi.org/10.1038/s41467-020-19188-9.
126. Zeighami A, Kern J, Yates AJ, Weber P, Bruno AAUS. West Coast droughts and heat waves exacerbate pollution inequality and can evade emission control policies. Nat Commun. 2023;14:1415. https://doi.org/10.1038/s41467-023-37080-0.
127. Chowdhury AFMK, Dang TD, Nguyen HTT, Koh R, Galelli S. The Greater Mekong's Climate-Water-Energy Nexus: How ENSO-Triggered Regional Droughts Affect Power Supply and CO2 Emissions. Earths Fut. 2021;9(3):2020–001814. https://doi.org/10.1029/2020EF001814.
128. Gómez C, Sanchez-Silva M, Dueñas-Osorio L, Rosowsky D. Hierarchical infrastructure network representation methods for risk-based decision-making. Structure and Infrastructure Engineering. 2013;9(3):260–74. https://doi.org/10.1080/15732479.2010.546415.
129. Zhou X, Duenas-Osorio L, Doss-Gollin J, Liu L, Stadler L, Li Q. Mesoscale modeling of distributed water systems enables policy search. Water Resour Res. 2023;59(5). https://doi.org/10.1029/2022WR033758.
130. Kazadi AN, Doss-Gollin J, Sebastian A, Silva A. Flood prediction with graph neural networks. In: Climate Change AI. Climate Change AI, 2022.
131. da Silva AL, Kocayusufoglu F, Jafarpour S, Bullo F, Swami A, Singh A. Combining physics and machine learning for network flow estimation. In: International Conference on Learning Representations, 2020.
132. Wong TE, Bakker AMR, Ruckert K, Applegate P, Slangen ABA, Keller K. BRICK v0.2, a simple, accessible, and transparent model framework for climate and regional sea-level projections. Geosci Model Dev. 2017;10(7),2741–2760. https://doi.org/10.5194/gmd-10-2741-2017.
133. Keller K, Helgeson C, Srikrishnan V. Climate risk management. Ann Rev Earth Planetary Sci. 2021;49(1):95–116. https://doi.org/10.1146/annurev-earth-080320-055847.
134. Bankes S. Exploratory modeling for policy analysis. Oper Res. 1993;41(3):435–49. https://doi.org/10/c7rgcr.
135. Brown CM, Ghile Y, Laverty M, Li K. Decision scaling: linking bottom-up vulnerability analysis with climate projections in the water sector. Water Resour Res. 2012;48(9). https://doi.org/10/gc3d7d.
136. Steinschneider S, McCrary R, Wi S, Mulligan K, Mearns LO, Brown CM. Expanded decision-scaling framework to select robust long-term water-system plans under hydroclimatic uncertainties. J Water Resour Plann Manag. 2015;141(11). https://doi.org/10.1061/(asce)wr.1943-5452.0000536.
137. Taner MÜ, Ray P, Brown C. Incorporating multidimensional probabilistic information into robustness-based water systems planning. Water Resour Res. 2019;55(5):3659–79. https://doi.org/10.1029/2018WR022909.
138. Lempert RJ. Robust Decision Making (RDM). In: Marchau VAWJ, Walker WE, Bloemen PJTM, Popper SW (eds) Decision making under deep uncertainty: from theory to practice. Cham: Springer International Publishing, (2019), pp. 23–51. https://doi.org/10.1007/978-3-030-05252-2_2.
139. Kasprzyk JR, Nataraj S, Reed PM, Lempert RJ. Many objective robust decision making for complex environmental systems undergoing change. Environ Model Softw. 2013;42:55–71. https://doi.org/10.1016/j.envsoft.2012.12.007.
140. Gupta SK, Rosenhead J. Robustness in sequential investment decisions. Manag Sci. 1968;15(2):18–229. https://doi.org/10.1287/mnsc.15.2.b18. arXiv:abs/2628850
141. de Neufville R, Smet K. Engineering Options Analysis (EOA). In: Marchau VAWJ, Walker WE, Bloemen PJTM, Popper SW, editors. Decision making under deep uncertainty: from theory to practice. Cham: Springer International Publishing; 2019. p. 117–32. https://doi.org/10.1007/978-3-030-05252-2_6.
142. Erfani T, Pachos K, Harou JJ. Real-options water supply planning: multistage scenario trees for adaptive and flexible capacity expansion under probabilistic climate change uncertainty. Water Resour Res. 2018;54(7):5069–87. https://doi.org/10.1029/2017wr021803.
143. Hino M, Hall JW. Real options analysis of adaptation to changing flood risk: structural and nonstructural measures. ASCE-ASME J Risk Uncertain Eng Sys A Civil Eng. 2017;3(3):04017005. https://doi.org/10.1061/ajrua6.0000905.
144. Quinn JD, Reed PM, Keller K. Direct policy search for robust multi-objective management of deeply uncertain socio-ecological tipping points. Environ Model Softw. 2017;92:125–41. https://doi.org/10.1016/j.envsoft.2017.02.017.
145. Schmidhuber J. Sequential Decision Making Based on Direct Search. In: Sun R, Giles CL (eds) Sequence learning: paradigms, algorithms, and applications. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2001, pp. 212–240. https://doi.org/10.1007/3-540-44565-X_10.
146. Sutton RS, Barto AG. Reinforcement learning: an introduction, Second. edition. Cambridge, Massachusetts and London, England: MIT Press; 2018.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.